



How Do Moral Emotions Shape Political Participation?

A Cross-Cultural Analysis of Online Petitions Using Language Models

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Data, Model

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Background

1) Definition of Moral Emotions

: Emotions motivated by concern for others and social justice, unlike general emotions

Type	Definition	Category
Moral Emotions	Other-condemning	Emotions that condemn others , such as anger, contempt, or disgust
	Other-praising	Emotions that praise others , such as admiration, gratitude, or awe
	Other-suffering	Emotions of empathy for the suffering of others , such as compassion or sympathy
	Self-conscious	Emotions that negatively evaluate oneself , such as shame, guilt, or embarrassment

2) Previous Research

- Evolutionary biology suggests that humans are drawn to moral emotion messages
- Moral emotions are key to the spread of political discourse online (moral contagion)
- They contribute to the formation and spread of political polarization

Data & Method

Allow students to be taken out of school for two weeks a year without penalty

Families face school fines for taking their children out of school to go on affordable holidays. This can be a particular issue for low-income families, and families with children that have additional needs, who want or need to avoid busier and more expensive periods.

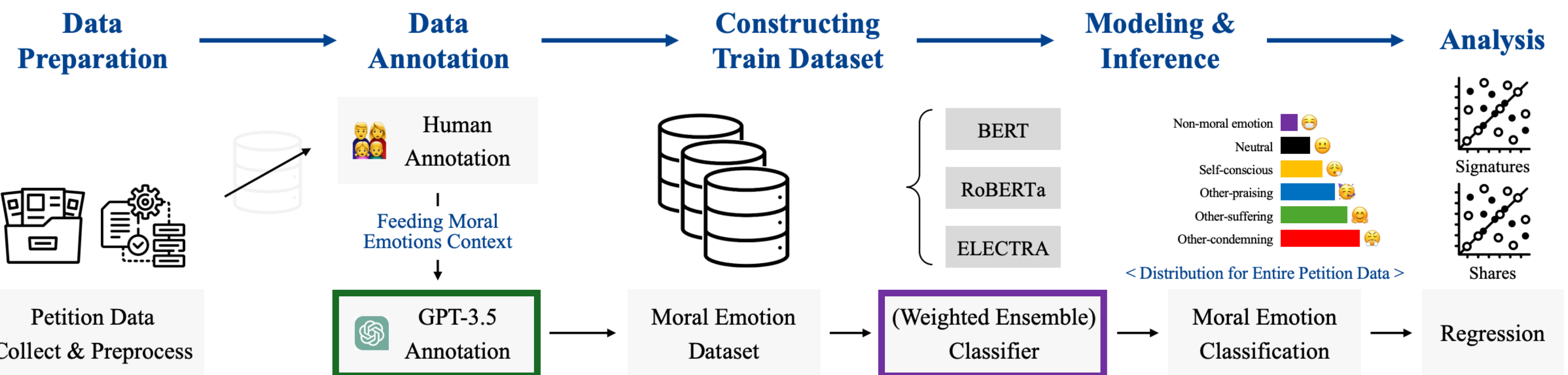
Sign this petition

249,047 signatures #Signature

Show on a map 100,000

Share this petition
#Shared on tweets

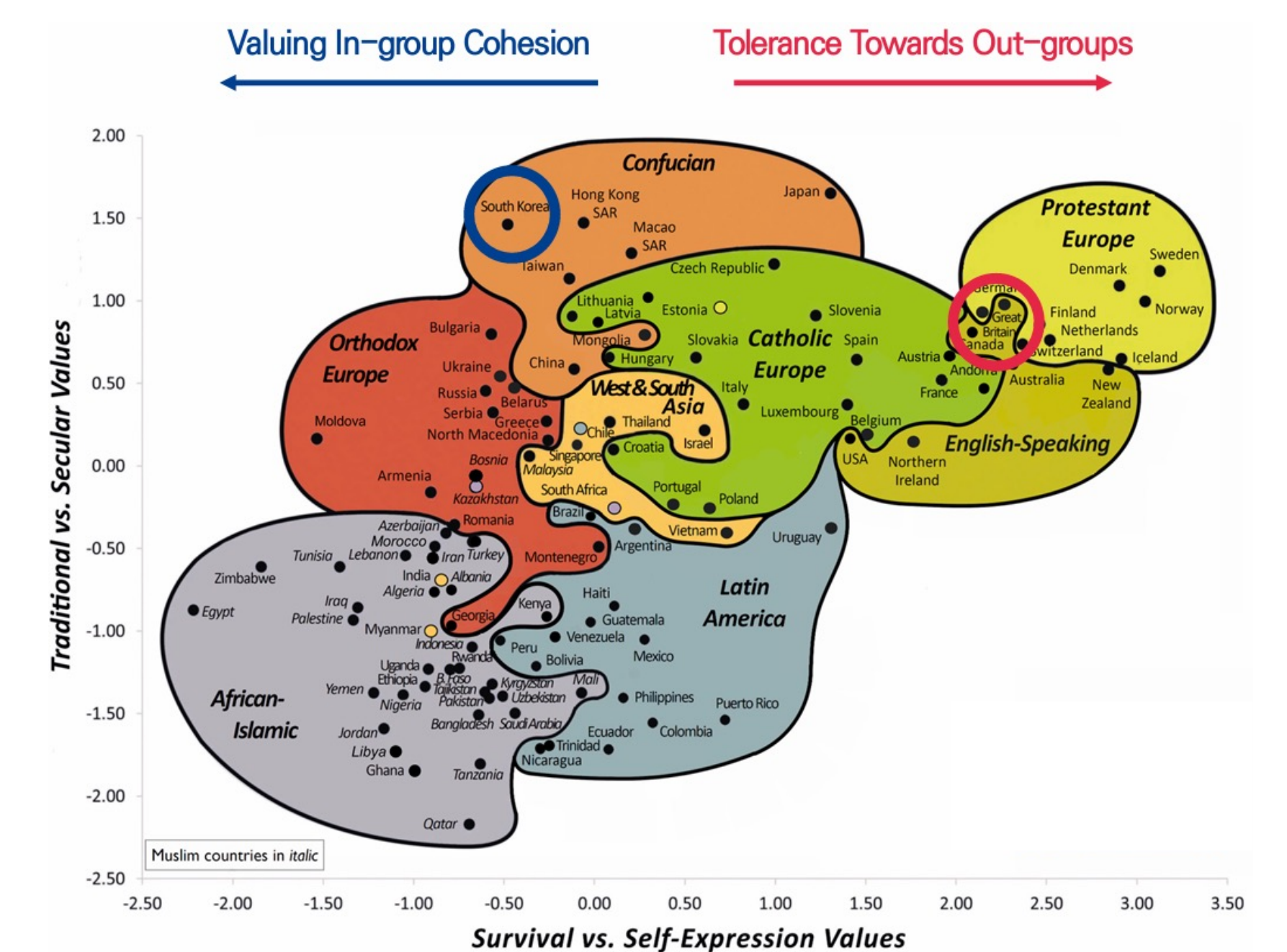
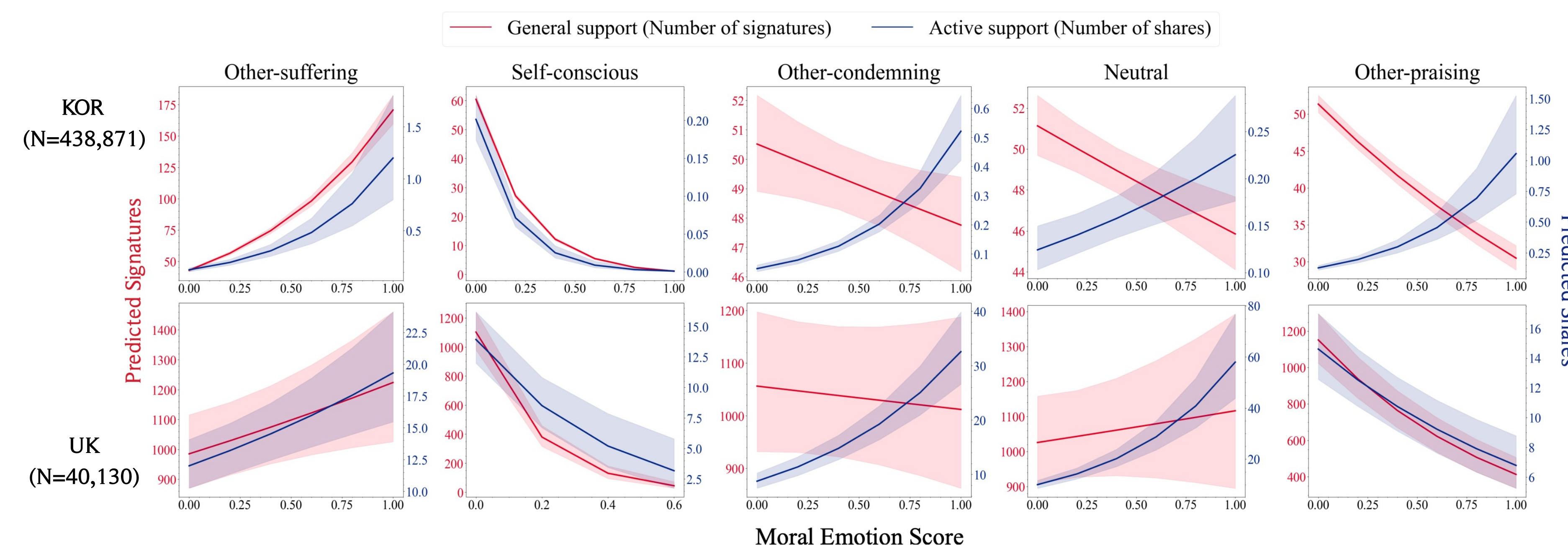
https://petition.parliament.uk/petitions/658365



	Train Size	Korean (KOR)		English (UK)	
		F1 (%)	Cost (\$)	F1 (%)	Cost (\$)
In-context Learning (GPT-3.5)	6	58.10	74.57	61.11	41.42
	12	58.25	104.87	62.23	57.98
	18	60.50	148.07	65.01	75.86
Fine-tuning (GPT-3.5)	200	85.30	338.17	73.48	164.10
	250	85.80	340.17	74.36	165.03
	300	86.78	342.16	74.26	165.93
Human Annotation	–	86.78	1,480.96	70.91	2,021.96

	Korean (KOR)	English (UK)
	Macro F1 (%)	
Fine-tuned GPT-3.5	86.78	74.36
BERT	88.58	67.60
RoBERTa	87.85	71.34
ELECTRA	89.14	75.23
Ensemble	89.50	73.67
Weighted Ensemble	89.78	75.36

Result & Discussion



1) Implication for Social Science

- Other-suffering Shapes Political Consensus
 - : Other-suffering fosters widespread consensus on petitions addressing social issues
- Other-condemning & Neutral Shape Political Polarization
 - : Political polarization can manifest in both emotional and rational forms
- Other-praising Impacts Vary Across Cultures
 - : Other-praising boosts in-group reputation and unity, leading to polarization
 - : In cultures valuing in-group cohesion like Korea, other-praising shapes polarization, while in those valuing out-group tolerance like the UK, it provokes antipathy

2) Implication for AI Community

- Cost-effective Analysis of Moral Emotions
 - : Fine-tuning GPT-3.5 and In-house modeling to cut annotation and inference costs
- Moral Emotion Classifier for Korean and English
 - : Moral emotion classifier for KOR, ENG, and extending to less-resourced languages
- Generative AI Risks (Political Persuasiveness)
 - : LLMs can understand and classify moral emotions, even from a limited of sentences
 - : These discoveries prompt future work into the potential of generative AI in crafting that may influence public opinion